Machine Learning–Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

A project-aligned empirical study of data preparation, exploratory analysis, K-means, and hierarchical clustering

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Abstract

This applied study evaluates the supplied Real Estate Buyer Segmentation project against its project requirements and saved analytical outputs. The source files contain 2,000 client records and 10,000 property listings, including 7,305 sold listings and 2,695 available listings. Sold records are reconciled to client IDs and aggregated into a one-row-per-buyer table of 2,000 records. The saved project's K-means solution uses nine standardized numeric features and assigns 628 buyers to High-Value Buyers (31.40%), 481 to Growth-Potential Buyers (24.05%), and 891 to Developing Buyers (44.55%). Recalculation reproduces the saved K=3 partition exactly (adjusted Rand index, ARI=1.000 against the saved labels), with silhouette 0.153756, Calinski–Harabasz 397.584, and Davies–Bouldin 1.941. The paper also performs an explicit Ward agglomerative hierarchical comparison and a sensitivity experiment that one-hot encodes categorical fields specified in the project requirements. On the nine numeric features, Ward clustering at K=3 yields silhouette 0.16168 but creates a 43-buyer cluster and a 1,079-buyer cluster; the saved K-means solution is more balanced and reproducible with the delivered labels. Adding categorical one-hot features changes the geometry and reduces the K=3 K-means silhouette to 0.10701, demonstrating sensitivity to feature representation. K=3 is therefore treated as the delivered business-facing segmentation, not as a universal statistical optimum. The analysis provides descriptive EDA and segment-level recommendations, while distinguishing the current dashboard implementation from additional controls requested in the project requirements.

Keywords: Real estate analytics; buyer segmentation; K-means; hierarchical clustering; one-hot encoding; exploratory data analysis; investment profiling; Streamlit

Project facts at a glance

Item	Verified value
Client source	2,000 rows × 12 columns; no missing cells or duplicate client IDs
Property source	10,000 rows × 9 columns; 7,305 sold; 2,695 available
Buyer-level output	2,000 rows; 2,000 unique client IDs; no missing cells
Saved model	K-means; K=3; nine standardized numeric features; random_state=42; n_init=50 audit
Reproduction check	Re-fit K=3 matches saved labels exactly (ARI=1.000)
K=3 saved-model metrics	Silhouette=0.153756; Calinski–Harabasz=397.584; Davies–Bouldin=1.941287
Data financial reconciliation	Sold sale-price sum = $2,520,750,960.84; equals buyer-level cumulative investment to cents

1. Introduction

Real-estate buyer populations can differ in purchase purpose, transaction history, property values, geography, financing indicators, and satisfaction. A buyer-level analytical view can help an organisation describe these differences consistently and formulate testable customer-engagement hypotheses. The project requirements document frames the task as financial analytics and real-estate market intelligence for Unified Mentor and Parcl Co. Limited, with an emphasis on buyer types, investment motivations, geographic variation, financing patterns, and more targeted recommendations.

This paper is grounded in the supplied project requirements document, source CSV files, saved buyer-level output, final segment profiles, business recommendations, and the current Streamlit application code. It separates (a) the saved segmentation that feeds the dashboard from (b) supplementary experiments run here to compare hierarchical clustering and categorical encoding. These are not silently treated as the same model.

2. Aim, objectives, and research questions

The aim is to document and evaluate the buyer-segmentation workflow against the project brief, report the observed profiles, and compare K-means with Ward agglomerative clustering under clearly stated feature representations.

Objectives: (1) audit the source tables and join coverage; (2) describe buyer and property data using EDA; (3) document the saved nine-feature K-means model; (4) compare candidate cluster counts using the elbow diagnostic and silhouette, Calinski–Harabasz, and Davies–Bouldin metrics; (5) conduct a Ward hierarchical comparison; (6) assess a PRD-aligned one-hot encoded feature representation; (7) interpret segments using geography, purpose, financing, demographics, and transaction metrics; and (8) identify dashboard and reproducibility gaps.

RQ1. Can the supplied client and property data be reconciled into a complete buyer-level table without missing or duplicate buyer identifiers? RQ2. What descriptive differences characterise the three saved segments? RQ3. How does K-means compare with Ward hierarchical clustering for K=2–6? RQ4. How sensitive are results to adding one-hot encoded categorical attributes? RQ5. Which requirements are represented in the current dashboard, and which remain to be implemented?

3. Data sources and data quality

The client source has 2,000 rows and 12 columns. The property source has 10,000 rows and 9 columns. The property status field contains 7,305 Sold records and 2,695 Available records. The client table has 0 missing cells, 0 exact duplicate rows, and 0 duplicated client IDs. The property table has 0 duplicated listing IDs. These are zero in the supplied files.

All sold records have a non-missing client reference, all parsed sold prices and dates are present, and every sold client reference matches a client_id in the client file. The saved buyer-level output contains 2,000 rows, 2,000 unique client IDs, and 0 missing cells. Sold transaction dates span 01 Jan 2024 to 01 Dec 2025; sale values range from $97,402.80 to $736,652.27. The sum of parsed sold sale prices is $2,520,750,960.84.

The Available records are not counted as completed buyer purchases because buyer attribution is absent for these records. The buyer-level analytical unit is one client, not one property row; this prevents buyers with multiple transactions from appearing multiple times in segment counts. The source does not establish the sampling frame, data-generation process, market representativeness, or economic currency provenance. Consequently, the results describe the supplied dataset only.

4. Exploratory data analysis

Buyer type is dominated by individuals (1,897; 94.85%), while companies account for 103 (5.15%). The acquisition-purpose field records Home for 1,385 buyers (69.25%) and Investment for 615 (30.75%). A loan was applied for by 736 buyers (36.80%).

The referral channel is Website for 1,103 buyers (55.15%), Agency for 705 (35.25%), and Client for 192 (9.60%). The United States is the most common recorded country (1,538; 76.90%). California is the most frequent region (633; 31.65%). Gender counts are 1,012 M and 988 F in the source labels.

These are sample descriptions, not population estimates. The strong concentration in the USA and California means the dataset is geographically imbalanced. Any geographic comparison should show denominators and avoid treating raw counts as rates. Country and region are also related variables; encoding both without feature weighting can give geography substantial influence over distances.

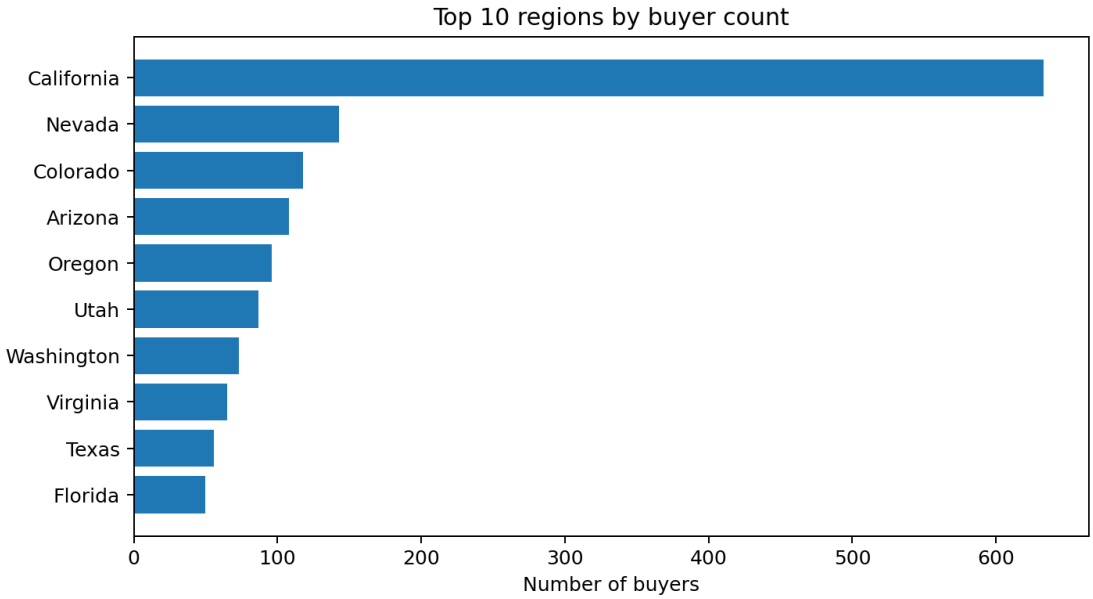


Figure 1. Ten most frequent regions by buyer count; raw counts reflect the supplied sample.

Table 1. Selected categorical distributions.

Variable	Category	Count	Share
Client type	Individual	1,897	94.85%
Client type	Company	103	5.15%
Acquisition purpose	Home	1,385	69.25%
Acquisition purpose	Investment	615	30.75%
Loan applied	Yes	736	36.80%
Referral channel	Website	1,103	55.15%
Referral channel	Agency	705	35.25%
Referral channel	Client	192	9.60%
Country	USA	1,538	76.90%

5. Feature engineering and preprocessing

The saved buyer-level artifact contains client attributes plus transaction-derived features. Transaction features include transaction_count, total_investment, average_property_value, median_property_value, minimum_property_value, maximum_property_value, total_floor_area_sqft, average_floor_area_sqft, unique_properties, first/last transaction dates, age, days_since_last_transaction, and investment_per_sqft. Age and recency in the saved pipeline use a fixed reference date of 31 December 2025; the recency value therefore ages as time passes.

The saved K-means model uses nine numeric variables: age, satisfaction_score, transaction_count, total_investment, average_property_value, minimum_property_value, maximum_property_value, days_since_last_transaction, and investment_per_sqft. These were standardised with StandardScaler. The model matrix does not directly include categorical variables. Identifiers and names are not clustering inputs.

To address the project brief's categorical-encoding requirement, a separate sensitivity experiment was performed using one-hot encoding for client_type, region, acquisition_purpose, referral_channel, country, loan_applied, and gender, combined with the same nine standardised numeric features. This produced an 87-column mixed feature matrix. It is reported as an additional experiment and does not replace the saved model used by the dashboard. One-hot encoding avoids imposing an arbitrary numeric order on nominal categories; however, the large number of region categories and the simultaneous use of country and region can alter distance weighting. A production mixed-data model should document feature weights and compare alternative representations. Nominal inputs were not ordinally label-encoded because assigning integers to categories would imply an order not established by the data; one-hot encoding was used for this sensitivity experiment.

6. Clustering methods and validation design

K-means partitions observations into K clusters by minimising within-cluster sum of squared distances to centroids. It requires K in advance and is sensitive to scale, initialization, and feature representation (MacQueen, 1967; scikit-learn developers, n.d.). The saved model uses K=3, random_state=42, and n_init=50 in the later audit. A re-fit with the documented nine numeric features reproduces the saved K=3 labels exactly (ARI=1.000).

Ward agglomerative clustering was used for the hierarchical comparison. It starts with individual observations and successively merges clusters; Ward linkage chooses merges that minimise increases in within-cluster variance. The experiment used the same scaled feature matrices and K=2–6. The 2,000-row dataset is small enough for this comparison on the current environment.

The elbow diagnostic records K-means inertia for K=2–10. Silhouette, Calinski–Harabasz, and Davies–Bouldin scores are reported for K=2–6. Higher silhouette and Calinski–Harabasz values and lower Davies–Bouldin values are generally preferred, but the criteria can disagree. The silhouette is defined from within-cluster cohesion and distance to the nearest alternative cluster; values near zero indicate overlap (Rousseeuw, 1987). Cluster sizes and agreement with the saved labels are reported alongside scores so that a high internal score does not hide a tiny cluster.

7. K-means results and selected buyer segments

The saved K=3 output partitions 2,000 buyers into three operationally named segments. Counts and shares reconcile to 2,000 and 100.00%. Segment labels are business interpretations of the output, not observed ground-truth buyer classes. The project requirements document lists four illustrative buyer archetypes, but the saved output contains three differently named operational segments; this paper does not relabel the saved clusters as those archetypes without supporting evidence. The table reports cumulative investment as the sum of buyer-level investment within a segment and mean investment per buyer separately.

Table 2. Recomputed segment profiles from the saved buyer-level output.

Segment	Buyers	Share	Cumulative investment	Mean invest./buyer	Mean property value	Mean tx.	Mean recency	Mean satisfaction
High-Value Buyers	628	31.40%	$953,614,517.27	$1,518,494.45	$353,259.07	4.33	169.1	3.22
Growth-Potential Buyers	481	24.05%	$681,772,224.50	$1,417,405.87	$435,537.92	3.25	234.7	2.94
Developing Buyers	891	44.55%	$885,364,219.07	$993,674.77	$294,993.80	3.39	231.1	2.94

Cumulative investment is a sum; mean investment per buyer is a mean. Currency provenance is not independently documented.

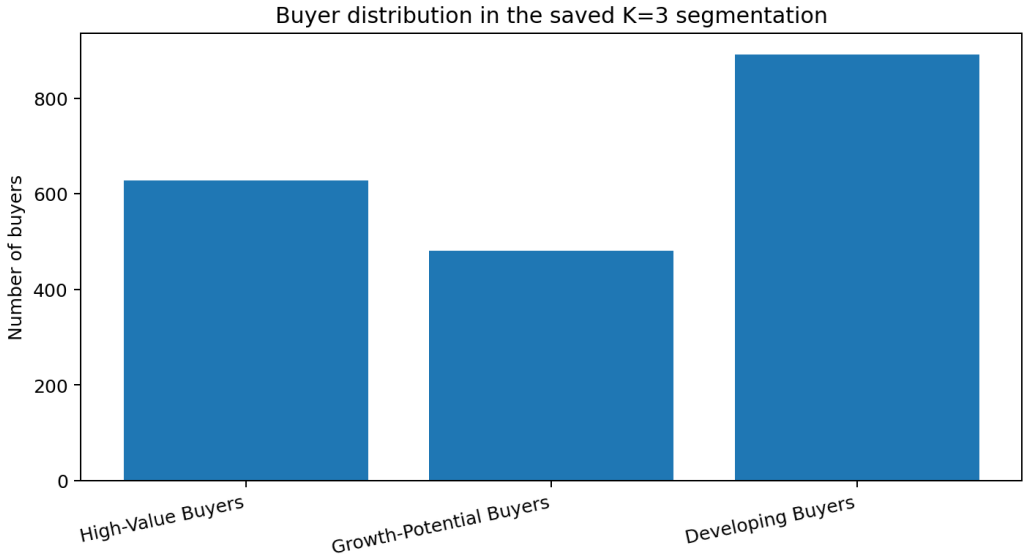


Figure 2. Buyer counts in the saved K=3 solution.

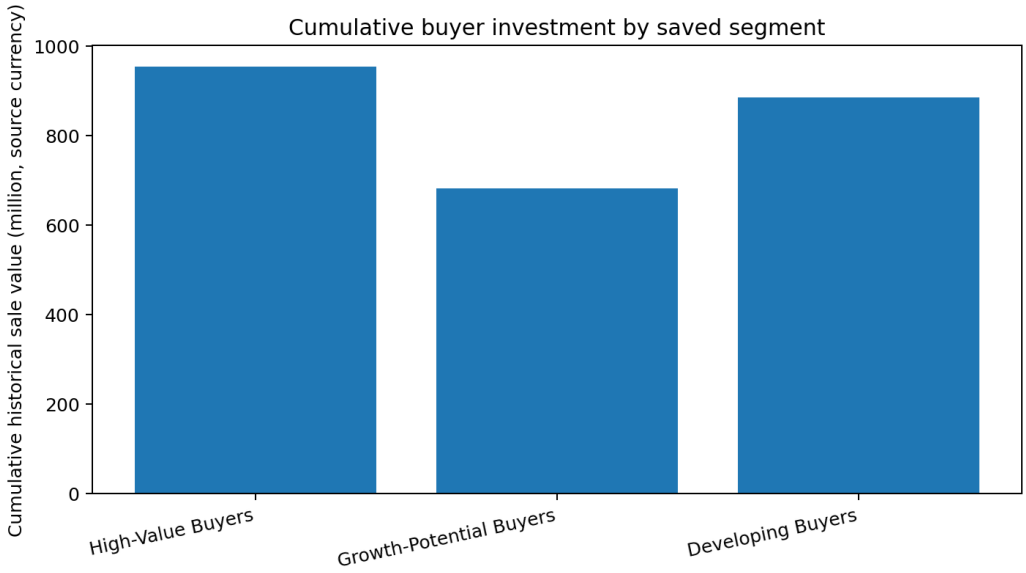


Figure 3. Cumulative historical sale value by saved segment; not profit.

8. Comparative clustering results

On the nine numeric features, K=2 produces a higher silhouette (0.18091) and Calinski–Harabasz score (502.043) than K=3. The elbow curve shows monotonically decreasing inertia, as expected when additional clusters are added; it does not by itself prove a unique optimum. The saved K=3 choice is therefore a practical business-facing choice, not the single best solution on every internal metric.

Ward hierarchical clustering on the same numeric features has silhouette 0.16168 at K=3, higher than K-means K=3 (0.15376), and a lower Davies–Bouldin value (1.65953 versus 1.94129). However, Ward produces a smallest cluster of only 43 buyers and a largest cluster of 1,079, and its partition has ARI 0.37754 against the saved labels. These differences show that cluster quality is multi-dimensional and algorithm-dependent; the hierarchical result is a comparison, not a drop-in replacement.

Adding categorical one-hot variables lowers the K=3 K-means silhouette to 0.10701 and changes the partition (ARI 0.95341 against the saved labels). Ward on the mixed matrix yields silhouette 0.09878, a smallest cluster of 37 and largest of 1,301. Because feature representation changes the geometry, these scores should only be compared within a clearly defined representation and not treated as a pure algorithm contest. The sensitivity analysis demonstrates that categorical encoding choices matter.

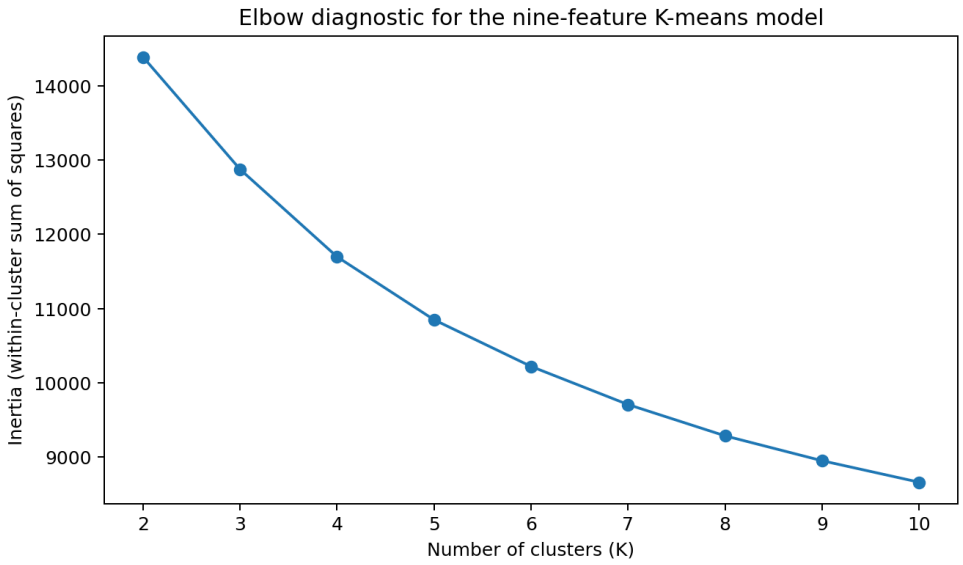


Figure 4. Elbow diagnostic for K=2–10.

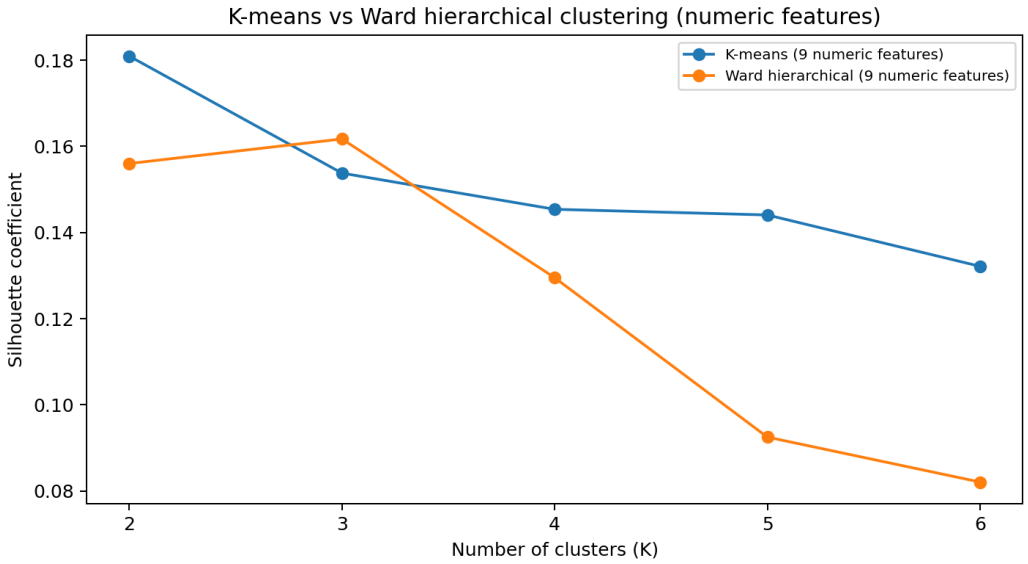


Figure 5. Silhouette comparison for K-means and Ward hierarchical clustering on numeric features.

Table 3. K=3 algorithm/representation comparison.

Model and representation	Silhouette	CH	DB	Min. cluster	Max. cluster	ARI vs saved
K-means; 9 numeric features	0.15376	397.584	1.94129	481	891	1.00000
Ward hierarchical; 9 numeric features	0.16168	332.621	1.65953	43	1,079	0.37754
K-means; numeric + one-hot	0.10701	262.648	2.43026	476	904	0.95341
Ward hierarchical; numeric + one-hot	0.09878	194.222	2.07212	37	1,301	0.17766

Table 4. Numeric K-means candidate diagnostics (n_init=20).

K	Inertia	Silhouette	Smallest cluster
2	14385.36	0.18077	941
3	12873.88	0.15376	481
4	11696.36	0.14536	50
5	10848.39	0.14374	50
6	10220.51	0.13227	48
7	9707.25	0.12969	49
8	9286.36	0.12593	49
9	8951.33	0.12242	42
10	8662.26	0.12201	42

9. Segment interpretation and business recommendations

High-Value Buyers (628; 31.40%) have the largest cumulative investment and the highest mean investment per buyer, mean property value, and mean transaction count among the saved segments. Their mean satisfaction is also higher than in the other two groups. The label describes historical observed activity; it does not prove high income, profitability, or future value.

Growth-Potential Buyers (481; 24.05%) have the highest mean property value and lowest mean transaction count, and their mean recency is the largest. This pattern can motivate a hypothesis of targeted follow-up, but it does not prove a buyer is likely to convert or grow in value.

Developing Buyers (891; 44.55%) form the largest group by count and have the lowest mean investment per buyer and mean property value. Their label is operational, not a statement about personal wealth or ability. The group may benefit from clearer property information or needs-based discovery, subject to consent and evaluation.

Recommended tests: (a) for High-Value Buyers, test retention and relevant premium-inventory recommendations; (b) for Growth-Potential Buyers, test preference-based follow-up and measure enquiry-to-sale conversion; (c) for Developing Buyers, test educational content and guided property discovery. Use controlled tests where feasible and measure incremental conversion, satisfaction, opt-outs, complaints, and repeat activity. These are hypotheses, not proven causal effects.

10. Streamlit dashboard and requirement alignment

The reviewed app.py reads existing buyer, segment-profile, and business-recommendation CSV outputs and does not retrain the model. The current interface provides buyer-segment multiselect filtering, buyer/segment/cluster KPIs, segment distribution, segment-profile and recommendation tables, a buyer explorer with country and region columns, and CSV export of the filtered buyer data.

The PRD names both One-Hot and Label Encoding. The saved K-means model does not use categorical inputs; the additional mixed-data experiment uses one-hot encoding only. Ordinal label encoding was not applied to nominal features because it would impose an arbitrary order, so

the literal encoding checklist is not fully implemented in the deployed model. The project requirements document also requests filters for country, region, acquisition purpose, and client type, plus a geographic buyer analysis that maps segments by region and a separate investor-behaviour view. Those specific controls and a geographic map are not present in the reviewed app.py. This is a traceable implementation gap rather than something the paper should claim has already been delivered. The current dashboard is a useful descriptive prototype, but it is not yet a full implementation of every PRD dashboard requirement.

11. Limitations, privacy, and reproducibility

The data do not include a verified ground-truth buyer class, income, credit capacity, profit margin, customer lifetime value, or campaign-response outcome. The clustering is unsupervised and descriptive; it cannot establish that segments improve revenue. The silhouette score for the saved K=3 model is modest, indicating overlap in the selected feature space. Financial variables are correlated and may dominate business interpretation even after standardisation.

The source does not document the sampling frame, provenance, currency basis, or permission for publication. The original client data include names, dates of birth, client IDs and other potentially identifying attributes. Raw client-level data and the final 29-column buyer table should not be made public without appropriate authorisation and privacy review. The paper reports aggregates only.

A reproducibility issue was identified in the older segment-summary notebook cell: it calculates a mean under a field called total_investment, while the corrected profile defines total_investment as a sum and average_buyer_investment as a mean. This manuscript uses the corrected definitions and verifies that segment sums reconcile to source sold-sale totals. Before formal publication, synchronise that notebook cell, rerun the entire raw-data-to-model pipeline from a clean environment, archive exact package versions, and regenerate outputs. The hierarchical and mixed-feature analyses in this paper are fresh diagnostic experiments on the saved buyer table; they do not mean the deployed dashboard uses those alternate models.

12. Conclusion

The project consolidates client and sold-property records into a buyer-level table and supplies a three-segment K-means output for descriptive real-estate market intelligence. The saved segmentation is reproducible from the documented nine numeric features and partitions 2,000 buyers into 628 High-Value, 481 Growth-Potential, and 891 Developing Buyers. The comparison experiments show that K-means and Ward hierarchical clustering produce different partitions, and that adding one-hot encoded categorical features changes internal scores and assignments. Therefore, the K=3 saved model is best described as a business-facing exploratory solution, not a statistically proven taxonomy.

The work fulfils the paper’s EDA, segment interpretation, investment profiling, and comparative clustering reporting objectives. The main next steps are to synchronise the summary notebook, rerun the full pipeline, implement the missing PRD dashboard filters/geographic visualisation, and validate proposed marketing actions with measured outcomes. No claim is made that the segments are causal, predictive, or representative beyond the supplied data.

13. References

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Appendix A. Reproducibility settings and feature definitions

Saved numeric feature set: age; satisfaction_score; transaction_count; total_investment; average_property_value; minimum_property_value; maximum_property_value; days_since_last_transaction; investment_per_sqft.

Numeric scaling: StandardScaler fitted to the nine numeric features in the saved buyer-level table.

Saved K-means validation settings: K=3; random_state=42; n_init=50 for the later candidate audit. Initial elbow screen used n_init=20.

Hierarchical validation: AgglomerativeClustering(n_clusters=K, linkage='ward') on the same scaled feature matrix; K=2–6.

Mixed sensitivity experiment: same nine scaled numeric features plus OneHotEncoder(handle_unknown='ignore', sparse_output=False) for client_type, region, acquisition_purpose, referral_channel, country, loan_applied, and gender; K-means and Ward evaluated for K=2–6.

Metrics: silhouette, Calinski–Harabasz, Davies–Bouldin, minimum/maximum cluster size, and ARI relative to the saved K=3 labels.

Age and recency reference date in the saved project: 2025-12-31.